

# Advancing Mental Health Research with Graph Neural Networks: A Comprehensive Survey

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**Abstract**—Graph Neural Networks (GNNs) have shown substantial potential in advancing mental health research, particularly in understanding disease mechanisms, improving diagnostic accuracy, predicting treatment outcomes, and enhancing public health strategies. GNNs excel at integrating diverse data types, such as neuroimaging, genetic, and clinical data, to model complex brain networks and disease relationships. This survey reviews the application of GNNs across five critical areas: (1) elucidating disease mechanisms, (2) diagnosis and classification of mental disorders, (3) risk prediction and intervention, (4) personalized medicine and treatment outcome evaluation, and (5) community prevention strategies. Despite challenges such as data heterogeneity, interpretability, and model scalability, GNNs offer promising solutions for more accurate, personalized, and dynamic mental health care.

**Index Terms**—Graph Neural Networks (GNNs), Mental Disorders, Disease Mechanisms, Diagnostic Classification, Diagnostic Classification, Risk Prediction, Personalized Medicine, Public Health Strategies, Neuroimaging, Multimodal Data Integration, Brain Networks, Mental Health Interventions

## I. INTRODUCTION

Mental disorders, including schizophrenia, depression, and bipolar disorder, significantly impact individual quality of life and present ongoing challenges for public health. Understanding the underlying mechanisms of these disorders and improving treatment outcomes have been central to research efforts. However, traditional methods face limitations in effectively integrating diverse data sources, which hinders the comprehensive understanding of these complex conditions. Graph Neural Networks (GNNs) provide an emerging approach by modeling complex, non-Euclidean relationships within brain networks and disease processes. This survey explores the role of GNNs in five key areas: (1) Disease Mechanisms—GNNs integrate neuroimaging, genomics, and clinical data to uncover underlying neurological disruptions; (2) Diagnosis and Classification—GNNs enhance diagnostic accuracy by capturing intricate patterns across multimodal data; (3) Risk Prediction and Intervention—GNNs model dynamic relationships to offer more precise risk assessment and timely interventions; (4) Personalized Medicine—GNNs optimize treatment strategies by modeling individualized drug responses and symptom progressions; (5) Community Prevention—GNNs help identify high-risk populations and improve resource allocation in community-level interventions. Despite challenges in data quality, model interpretability, and scalability, GNNs hold transformative potential for mental health research and practice.

## II. BACKGROUND

### A. Mental disorders

Mental health disorders are a major global concern, affecting the overall well-being of society and the quality of an individual's life. Based on symptomatology and underlying mechanisms, these disorders can be classified into five major categories [1], [2]:

1. Bipolar Disorders: Includes major depressive disorder (MDD) and bipolar disorder, involving emotional dysregulation. Depression leads to persistent sadness and cognitive impairments, while bipolar disorder alternates between manic and depressive episodes, significantly affecting daily life [1], [2].

2. Schizophrenia: Primarily characterized by hallucinations, delusions, cognitive deficits, and social withdrawal [1]. It often emerges in early adulthood, requiring long-term management due to persistent symptoms [2].

3. Anxiety and Trauma: Covers generalized anxiety disorder (GAD), panic disorder, PTSD, and social anxiety disorder, all marked by excessive fear, worry, or distress. PTSD, linked to traumatic experiences, often leads to emotional numbness and risk-taking behaviors [1], [2].

4. Neurodevelopmental Disorders: Prevalently in ADHD, characterized by inattention, hyperactivity, and impulsivity. While often diagnosed in childhood, symptoms frequently persist into adulthood [2].

### B. CMH Data

GNNs leverage diverse data to uncover functional and structural brain abnormalities. These data come from modalities like fMRI, electroencephalogram (EEG), sMRI, fNIRS, PET, and clinical information (e.g., questionnaires, medical history, genetic data), collectively known as Cognitive and Mental Health Data (CMH Data). These multidimensional datasets provide complementary insights into the neuropathology of mental disorders, with studies highlighting the importance of multimodal data fusion in revealing brain network abnormalities [3].

Among these, fMRI and EEG are widely used for their complementary strengths. fMRI [4] offers high spatial resolution to identify disrupted functional connectivity, while EEG [5] provides high temporal resolution to capture dynamic brain activity. Together, they enable GNN-based mental disorder analysis, offering a comprehensive view of brain function through multimodal integration.

### C. Graph Neural Network Techniques

- 1) *Fundamental Concepts of GNNs*: GNNs [6] are designed to handle graph-structured data, where a graph

$G = (V, E)$  consists of a set of nodes  $V$  and a set of edges  $E$ . In most GNN models, each node's representation is iteratively updated under a message passing framework by aggregating information from its neighbors. A general update equation is given by:

$$h_i^{(l+1)} = \text{UPDATE}^{(l)}(h_i^{(l)}, \text{AGGREGATE}^{(l)}(\{h_j^{(l)} : j \in \mathcal{N}(i)\})), \quad (1)$$

where  $\mathcal{N}(i)$  denotes the set of neighbors of node  $i$  and  $h_i^{(l)}$  represents the feature of node  $i$  at layer  $l$ .

In practice, a linear transformation followed by a non-linear activation is often employed:

$$h_i^{(l+1)} = \sigma(W^{(l)} \cdot \text{AGGREGATE}^{(l)}(\{h_j^{(l)} : j \in \mathcal{N}(i)\})), \quad (2)$$

where  $W^{(l)}$  is the weight matrix for layer  $l$  and  $\sigma(\cdot)$  denotes a non-linear activation function.

Below, we introduce several representative GNN models and their variants in the order of their publication.

2) *Graph Convolutional Networks (GCN)*: GCN [7] is a classical method that performs normalized convolution on graphs. The idea is to extend the adjacency matrix with self-loops and then normalize it for local feature aggregation. First, the self-loop extended adjacency matrix is defined as:

$$\tilde{A} = A + I, \quad (3)$$

where  $I$  is the identity matrix.

Based on this, the layer update for GCN is defined as:

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}), \quad (4)$$

where  $\tilde{D}$  denotes the degree matrix of  $\tilde{A}$ ,  $H^{(l)}$  represents the node features at layer  $l$ , and  $W^{(l)}$  is the trainable weight matrix.

3) *Graph Attention Networks (GAT)*: GAT [8] introduces an attention mechanism that allows the model to assign different levels of importance to its neighbors when aggregating information. For each node  $i$  and its neighbor  $j$ , the attention coefficient is computed as:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))}, \quad (5)$$

where  $\parallel$  denotes vector concatenation.

Then, the node feature is updated using the attention coefficients:

$$\mathbf{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j\right). \quad (6)$$

4) *GraphSAGE*: GraphSAGE [9] proposes an inductive framework that generates node embeddings by sampling and aggregating information from the node's neighborhood. Its general update rule is:

$$\mathbf{h}_v^{(k)} = \sigma\left(\mathbf{W}^{(k)} \cdot \text{AGGREGATE}^{(k)}(\{\mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v)\} \cup \{\mathbf{h}_v^{(k-1)}\})\right). \quad (7)$$

For example, when using a mean aggregator, it can be expressed as:

$$\text{AGGREGATE}^{(k)} = \text{mean}(\{\mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v)\}). \quad (8)$$

5) *Graph Isomorphism Networks (GIN)*: GIN [10] addresses the discriminative power for graph isomorphism tests by proposing a new design. Its update formula is given by:

$$\mathbf{h}_v^{(k)} = \text{MLP}^{(k)}\left((1 + \epsilon^{(k)}) \mathbf{h}_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)}\right), \quad (9)$$

where  $\text{MLP}^{(k)}$  is a multi-layer perceptron at the  $k$ th layer, and  $\epsilon^{(k)}$  is either a learnable parameter or a fixed constant. An equivalent aggregation view is:

$$\mathbf{h}_v^{(k)} = \text{MLP}^{(k)}\left(\text{AGGREGATE}^{(k)}(\{\mathbf{h}_v^{(k-1)}\} \cup \{\mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v)\})\right). \quad (10)$$

6) *Graph Convolutional Neural Networks (GCNN)*: In GCNN [11] designs, spectral filters based on Chebyshev polynomials are often used to approximate the spectrum of the graph Laplacian. The output is expressed as:

$$y = \sum_{k=0}^K \theta_k T_k(\tilde{L}) x, \quad (11)$$

where  $T_k(\tilde{L})$  denotes the Chebyshev polynomial. Its recursive formula is given by:

$$T_k(\tilde{L}) = 2\tilde{L}T_{k-1}(\tilde{L}) - T_{k-2}(\tilde{L}), \quad T_0(\tilde{L}) = I, \quad T_1(\tilde{L}) = \tilde{L}. \quad (12)$$

Here,  $\tilde{L}$  is the normalized graph Laplacian.

7) *Ci-GNN*: Ci-GNN [12] (Contextual and Interpretable GNN) enhances the interpretability and performance of node representations by incorporating contextual information. Its node update can be expressed as:

$$h_i' = \sigma\left(W_1 h_i + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} (W_2 h_j + W_3 c_{ij})\right), \quad (13)$$

where  $c_{ij}$  represents the contextual information between nodes  $i$  and  $j$ . The corresponding attention coefficient is computed as:

$$\alpha_{ij} = \frac{\exp(\beta^T [h_i \parallel h_j \parallel c_{ij}])}{\sum_{k \in \mathcal{N}(i)} \exp(\beta^T [h_i \parallel h_k \parallel c_{ik}])}. \quad (14)$$

Here,  $\beta$  is a learnable parameter vector and  $\parallel$  denotes vector concatenation.

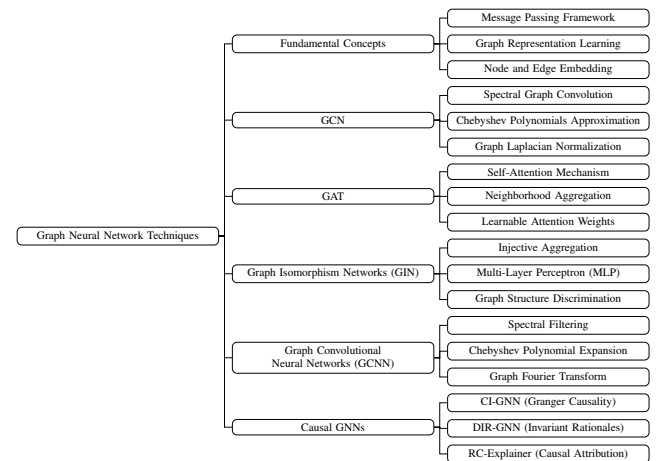


Figure 1. Graph Neural Network Techniques

### III. GNNs IN MENTAL DISORDERS MECHANISMS

#### A. Introduction

GNNs play a critical role in elucidating mental disorders mechanisms and integrating multimodal data in mental health research. By modeling complex brain connectivity and utilizing neuroimaging, GNNs provide insights into both healthy and pathological brain states [13]–[15]. These models are particularly valuable in understanding the underlying mechanisms of psychiatric disorders, including the dynamic interactions within neural networks. However, challenges such as interpretability, data scarcity, and effective multimodal integration limit the full potential of GNNs in advancing psychiatric applications [13], [14], [16].

#### B. Existing Methods

Traditional methods for understanding mental disorder mechanisms include genomics, neuroimaging (e.g., MRI, fMRI), clinical data, and behavioral analysis [17], [18]. While valuable, these approaches often fail to effectively integrate multimodal data, leading to lost information and overlooked relationships. Genomics links gene variations to disorders, while neuroimaging identifies brain abnormalities [17]. Studied in isolation, these methods cannot fully explain the complex mechanisms of mental disorders [19]. Recent efforts focus on multimodal integration for a more comprehensive view of mental health's biological basis [20].

However, data integration remains challenging due to disorder heterogeneity and data complexity [21], requiring further refinement [22].

#### C. Advantages and Limitations of Existing Methods

Traditional methods for elucidating mental disorder mechanisms, such as genomics, neuroimaging (e.g., MRI, fMRI), clinical data, and behavioral analysis, offer valuable insights. Genomics identifies genetic variations, and neuroimaging reveals brain abnormalities [23], [24]. Clinical and behavioral data help understand symptoms and disease progression [24], [25]. However, these approaches often fail to integrate multimodal data, resulting in fragmented insights and missed relationships [26], [27].

Despite their effectiveness in isolating individual data sources, traditional methods struggle to synthesize multimodal information, particularly when genomic and neuroimaging data, which could complement each other, are analyzed separately [24], [28].

As mental health research advances, integrating multimodal data is crucial for a comprehensive understanding of complex disorders like schizophrenia, depression, and bipolar disorder [24], [27]. This highlights the need for more integrated models [29].

#### D. GNNs in Mental Disorders Mechanism Integration

Traditional methods for elucidating mental disorders mechanisms focus on individual data sources, such as genomics, neuroimaging, or clinical data, often failing to integrate them, leading to missed relationships and incomplete understanding [30], [31]. These approaches rely on separate models, hindering unified analysis [32].

GNNs integrate heterogeneous data through graph-based modeling, processing genomic, imaging, clinical, and behavioral data in a unified structure [33]. By representing genes, brain regions, and symptoms as nodes, GNNs capture relationships like gene-disease associations and brain region connectivity, providing a comprehensive view of mental disorders mechanisms [34], [35].

GNNs excel in handling multi-omics data, essential in cancer and neurodegenerative research, revealing complex links between molecular mechanisms and clinical manifestations [30], [33], [35]. In contrast, traditional methods fail to capture these interdependencies [31]. Approaches like MODIG and Net-DMPred, integrating multi-omics data, offer a holistic view of mental disorders mechanisms [36].

#### E. Applications of GNN in the Mechanisms of Psychiatric Disorders

a) *MDD*: GNNs significantly advance understanding of MDD mechanisms, with methods like DepressionGraph, topology-optimized GNNs, and multi-site GCNs offering advantages in brain connectivity analysis.

DepressionGraph integrates node features and graph topology in a dual-channel GNN, capturing both local and global information. Its Transformer-Encoder structure models dynamic time-series data, improving brain network dynamics analysis. In a 533-subject dataset, it outperformed traditional models like SVM and RF, with an F1-Score of 71.13% [37].

Topology-optimized GNNs focus on shortest paths in the brain network, sparsifying the graph to retain only 2.5% of the original connections, boosting accuracy to 93% and reducing biases [38].

A multi-site GCN study using data from 16 sites identified MDD-related brain regions with 81.5% accuracy, showing stable performance across diverse datasets [39].

Benchmarking of various GNN models for fMRI analysis showed the following:

Table I  
BENCHMARKING OF VARIOUS GNN MODELS FOR FMRI ANALYSIS.

| Benchmarking of GNN Models |              |                 |                 |
|----------------------------|--------------|-----------------|-----------------|
| GNN Models                 | Accuracy (%) | Sensitivity (%) | Specificity (%) |
| GCN                        | 62.6         | 57.6            | 67.8            |
| GIN                        | 62.7         | 63.3            | 58.1            |
| GAT                        | 62.9         | 69.5            | 56.4            |
| ST-GCN                     | 58.2         | 48.6            | 67.8            |
| AST-GCN                    | 60.7         | 44.6            | 75.8            |

These results highlight GNNs' superior classification over traditional methods.

b) *Schizophrenia*: GNNs provide superior insights into the pathophysiological mechanisms of schizophrenia, outperforming traditional methods in brain connectivity analysis. GNNs applied to EEG data revealed significant MMN deficits in first-episode schizophrenia (FESZ) and chronic schizophrenia (CSZ) patients, achieving 85.8% accuracy, compared to SVM's 80.92% [40].

GCNs applied to resting-state fMRI data from 505 schizophrenia patients and 907 controls achieved 85.8%

accuracy and an AUC of 0.926, identifying dysconnectivity in the putamen, pallidum, and caudate. GCNs outperformed SVM (80.9% accuracy, AUC 0.897) by dynamically learning complex brain network patterns and correlating nodal efficiency with symptom severity [41].

Research on GABAergic neurons highlights reduced GAD67 expression in the prefrontal cortex, impairing cognitive functions such as working memory and attention. GNNs effectively model these disruptions, providing new insights into schizophrenia's cognitive deficits [42].

*c) Bipolar disorder:* GNNs, such as DynBrainGNN and BrainNN, offer significant advantages over traditional machine learning methods in studying bipolar disorder. Traditional approaches like SVM, logistic regression, and random forests rely on static feature extraction, often missing dynamic brain network relationships. DynBrainGNN utilizes dynamic functional connectivity (FC) to analyze brain network fluctuations, identifying abnormalities in mood-related regions. In the ABIDE dataset, it achieved 72% accuracy and an F1 score of 0.72 [43]. By capturing temporal dynamics, it outperformed traditional methods in the SRPBS dataset with 83% accuracy and an F1 score of 0.89 [43]. BrainNN integrates structural and functional brain network data using contrastive learning, improving accuracy by 11% over traditional methods in bipolar disorder and HIV datasets [16]. GNNs also model temporal changes in emotional states, capturing their effect on brain connectivity, a feature overlooked by traditional methods [44].

#### F. Current Challenges and Future Directions

*a) Data Imbalance and Sparse Labeling:* Data imbalance and sparse labeling are key challenges in biomedical tasks like disease prediction and gene-disease association. Imbalance occurs when positive samples (e.g., diseased individuals) are outnumbered by negative samples (e.g., healthy individuals), while sparse labeling refers to limited labeled data.

Zafeiropoulos et al. emphasized GNNs' ability to capture complex relationships despite data imbalance [45]. Mastropietro et al. addressed this by using Positive-Unlabeled (PU) learning, leveraging positive labels and pseudo-label propagation to improve performance with sparse data [46].

KGNMDA enhances microbe-disease association prediction by incorporating Gaussian kernel similarity features into GNNs for sparse-labeled contexts [47]. GNNExplainer improves model interpretability by identifying key biomarkers crucial for AD progression [48].

*b) Noisy Data and Data Quality Issues:* Noise and data quality issues pose significant challenges in computational models, particularly in biomedical and epidemic research. Sun et al. used GCNs with GAT to mitigate noise and improve accuracy in metabolite-disease prediction [49]. Shi et al. applied a heterogeneous Graph Neural Network (HGNNLDA) for lncRNA-disease associations, using attention-based aggregation to filter noise [50].

In epidemic modeling, Liu et al. noted the limitations of mechanistic models due to noise from factors like mobility and interventions [51]. They proposed combining GNNs

with traditional models to better handle noisy data. Khemani et al. introduced a Granger Causality-inspired GNN (CI-GNN) to reduce spurious correlations in neuroimaging data [52].

*c) Interpretability of Models and Inference of Causality:* Interpretability and causal inference are essential in GNNs, particularly in healthcare and bioinformatics. GNNs model complex relationships through message passing between nodes [53], but their complexity often hampers interpretability. GAT enhance transparency by assigning weights to neighbors, revealing the influence of specific nodes on predictions [54], [55].

In ncRNA-disease prediction, GNNs capture high-order dependencies via multi-layer feature propagation, with semi-supervised learning addressing sparse supervision [56]. Causal inference methods, such as Granger CI-GNN, identify causally relevant subgraphs, improving model explainability [53].

*d) Cross-Domain Adaptation Challenges:* Cross-domain adaptation refers to the decline in model performance when applied to data from different domains or devices due to distribution shifts. This is a prominent issue in brain connectivity studies, particularly when using GNNs to analyze neuroimaging data from diverse sources or scanning techniques, such as MRI, fMRI, and DTI [13], [57].

Mohammadi and Karwowski note that while GNNs are effective at capturing complex relationships, they struggle with integrating multimodal data due to distributional differences [13]. Zoppi et al. emphasize that while multi-omics integration can improve model adaptability, challenges such as sparse data and missing labels persist [21], [57].

*e) Computational Complexity and Resource Demands:* GNNs face significant challenges in computational complexity and resource demands, particularly with large-scale graphs. Redundant computations, accounting for 92.4% of total operations, lead to high memory usage and inefficiencies [58].

To address these issues, Zhou et al. introduced the GNNear accelerator, utilizing Near-Memory Processing (NMP) to optimize memory access and improve efficiency [59]. However, irregular graph sparsity still leads to resource underutilization. Shivdikar et al.'s NeuraChip accelerator decouples multiplication and accumulation operations, achieving up to 22.1x speedup [60]. Additionally, Huang et al. developed CATGNN, a distributed system that scales GNNs to large graphs through dynamic partitioning [61].

In summary, advanced solutions such as NMP, specialized accelerators, and distributed training are essential for enhancing the scalability and efficiency of GNNs [61].

## IV. DIAGNOSIS AND CLASSIFICATION OF MENTAL DISORDERS

### A. Introduction

Early diagnosis and precise classification in mental health remain challenging. Traditional methods, based on subjective judgments, suffer from inconsistent criteria and limited biological insight [62]. Advanced computational

approaches for multimodal data analysis offer a promising solution.

Multimodal data—like brain imaging (e.g., fMRI, sMRI, PET), EEG, and clinical information—enable biomarker extraction and graph-based brain interaction modeling. Traditional methods struggle with non-Euclidean graph data, missing local and global patterns [3]. GNNs, by integrating node and connectivity information, excel in early diagnosis and precise classification [63].

### B. Existing Methods

1) *Fundamental Machine Learning Techniques:* Early mental disorder diagnosis used manual feature engineering with brain imaging, neurophysiological data, and clinical assessments.

SVMs were widely used. Yang et al. [64] achieved 87% accuracy in schizophrenia diagnosis by integrating fMRI and SNP data. Schnack et al. [65] reached 90% accuracy in classifying schizophrenia, bipolar disorder, and controls using gray matter density. Sacchet et al. [66] classified MDD via diffusion imaging and graph metrics, while Karstoft et al. [67] improved PTSD predictions by combining neurobiological and clinical features.

To improve robustness, ensemble methods like random forests and boosting algorithms were introduced. Greenstein D et al. [68] used random forests on brain, clinical, and genetic data to predict disease. Maenner M J et al. [69] applied random forests to classify ASD with cross-year accuracy.

2) *Deep Learning Techniques:* Since the 2010s, deep learning methods like CNNs and RNNs have advanced feature learning for mental disorder datasets, enhancing pattern representation.

CNN extracts spatial features from 2D/3D brain imaging data. Acharya U. R. et al. [70] used CNN to learn EEG features for distinguishing depressed patients from healthy controls. Zhang P. et al. [71] proposed an R3DCNN framework combining recurrent and 3D convolutional networks to classify mental workload levels. L. Zou et al. [72] applied 3D CNN to MRI scans, achieving 69.15% accuracy, outperforming other classifiers with limited training samples.

RNN captures temporal dependencies in time-series data like EEG and clinical records. R. Supakar et al. [73] used RNN-LSTM for schizophrenia diagnosis from EEG signals. Penchina B. et al. [74] applied LSTM to identify anxiety states from continuous EEG signals, achieving 90.82% accuracy.

Additionally, There also have studies that explored autoencoders, VAEs, and transfer learning to address labeled data scarcity and achieve higher-level feature abstraction [75].

### C. Strengths and Weaknesses of Existing Methods

Logistic regression is efficient and interpretable but struggles with nonlinear relationships and complex brain networks, limiting its use for high-dimensional neuroimaging. SVMs handle high-dimensional data well with kernel functions but depend heavily on kernel selection and parameter tuning.

Deep learning methods like CNNs excel at extracting local spatial features from imaging data (e.g., fMRI, sMRI). Yet, traditional CNNs are designed for Euclidean data and cannot represent the non-Euclidean interconnectivity of brain regions. RNNs, on the other hand, effectively model temporal dependencies, capturing dynamic changes in behavior and neural activity. However, they struggle to represent complex spatial relationships, limiting their ability to reflect global brain network structures [76]. Integrating multimodal data, such as neuroimaging, genetic, and clinical data, remains a key challenge, as current methods lack effective mechanisms for combining these diverse sources.

### D. Bridging the Gap with GNN Methods

GNN excels with non-Euclidean data, capturing local and global brain network features. Unlike SVM, CNN, and RNN, GNN models complex relationships and topologies, revealing latent connectivity and brain structures.

GNN integrates node (brain metrics) and edge (connectivity) information, enhancing brain network representation. Additionally, GNN excels in scalability and multimodal integration. Combining GNN with CNN enables joint learning on neuroimaging and clinical data, addressing heterogeneity and nonlinear relationships, and improving diagnostic accuracy.

In summary, GNN overcomes existing limitations by capturing complex neural networks and integrating multimodal data. Unlike deep learning models limited to Euclidean structures, GNN provides more intuitive and interpretable biomedical insights [63].

### E. GNN Applications in Mental Health Disorders

GNNs have become a focus in mental disorder diagnosis, especially for depression and schizophrenia, expanding from imaging data (fNIRS, fMRI, EEG) to non-imaging data (text, speech) and leveraging multimodal fusion to analyze brain connectivity. Models have evolved from basic GNNs to advanced techniques such as self-attention pooling and graph autoencoders. Below, the content highlights key studies based on innovation, citations, and publication date, summarizing GNN advancements in mental health diagnosis.

1) *GNN-Based Approaches for Depression Diagnosis and Classification:* In this module, I selected nine papers covering the progression from initial explorations of single modalities to multimodal information integration and multi-stage graph fusion techniques. The diagram below provides an overview and summary framework, highlighting the evolution of methods and technologies across these studies.

Some studies use fNIRS and fMRI data to construct functional connectivity matrices, capturing spatiotemporal and topological features. Q. Yu et al. [77] integrated time-domain statistics with spatial correlations to improve diagnostics. E.N. Pitsik et al. [38] showed that aligning GNN depth with network topology and sparsifying inputs enhanced classification. Venkatapathy et al. [78] employed an ensemble of GNNs (e.g., GCN, GAT, GraphSAGE) to capture brain network complexity and distinguish subtypes. These studies highlight the effectiveness of diverse

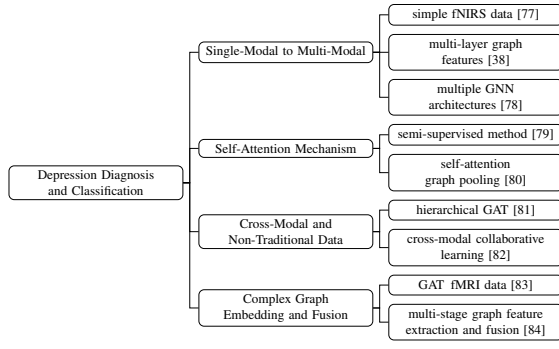


Figure 2. Overview of Key Works in GNN-Based Methods for Depression Diagnosis and Classification

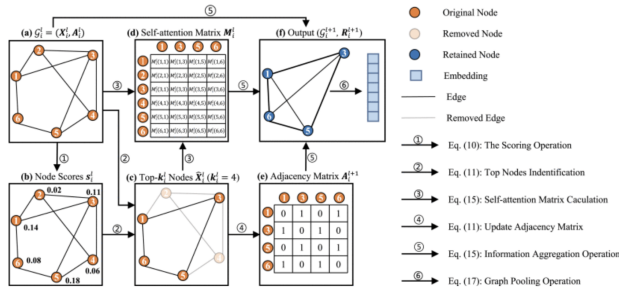


Figure 3. A schematic diagram of the self-attention graph pooling module, excerpted from T. Chen et al. (2022) [80].

graph-based strategies in improving robustness and accuracy.

In the application of EEG data, some studies have proposed solutions to address the issue of insufficient labeled data. D. Wang et al. [79] used a semi-supervised self-training strategy with a self-organizing incremental neural network and GCN, improving label propagation and generalization. T. Chen et al. (2022) [80] applied multi-attention graph pooling and soft label strategies to refine graph representations and reduce information loss, enhancing feature discrimination.

The method uses a three-stage hierarchical optimization—node importance evaluation, connection sparsification, and attention enhancement—to model brain network dynamics. A self-attention graph pooling mechanism with 11 norm quantifies node contributions, selects top-k nodes, and prunes redundant connections via sparsemax for noise suppression. A bidirectional attention module incorporates brain region priors, while residual connections preserve topology to prevent feature loss and isolated nodes.

Another research direction explores non-traditional data like text and speech for multimodal integration. M. Niu et al. [81] used a hierarchical context-aware graph attention model to capture text-speech relationships in clinical interviews, achieving high-precision assessments. T. Chen et al. (2023) [82] introduced a novel GNN framework for efficient cross-modal fusion by extracting shared and specific features for each modality. The framework uses GNN to extract common features across modalities, constrained by similarity loss to ensure consistency among shared features. At the same time, independent GNN branches extract unique features for each modality, with a decorrelation loss enforcing orthogonality between shared and

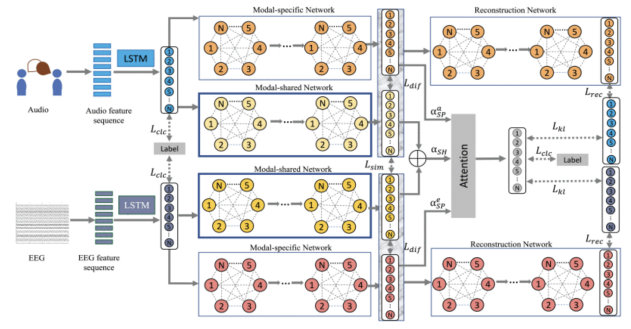


Figure 4. A schematic diagram of the multimodal MDD detection framework, excerpted from T. Chen et al. (2023) [82].

specific features, reducing redundancy while preserving inter-modal heterogeneity.

Finally, Learnable attention weights fuse shared and modality-specific features, enhancing multimodal integration and diagnostics. F. Noman et al. [83] proposed a graph autoencoder to learn compact embeddings, capturing non-Euclidean features to improve accuracy. Y. Kong et al. [84] combined deep subspace learning with graph convolution for better sensitivity and interpretability. The former emphasizes embedding compactness, while the latter focuses on richer cross-stage integration, advancing graph-based diagnostics.

In summary, GNN-based methods for depression diagnosis model complex graphs and multimodal data, enhancing sensitivity to brain abnormalities and enabling cross-domain integration. These advancements support precise diagnosis, automated assessment, and future research.

2) *Applications of GNNs in Schizophrenia:* This module focuses on research utilizing GNNs for the diagnosis and classification of schizophrenia.

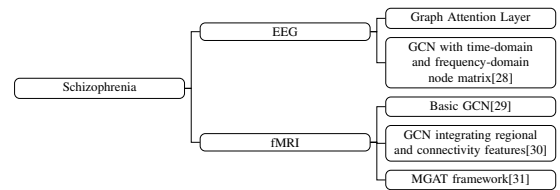


Figure 5. Summary of GNN-based methods for schizophrenia diagnosis and classification.

Research on EEG data lays the groundwork in this field. Chan C et al. [85] used 250ms of EEG data from 8 channels to construct brain connectivity graphs and designed two GATs to capture local and global connectivity, proving the feasibility of classification with simple data. Yin G et al. [86] constructed a node matrix from EEG data of 103 patients and 92 controls, incorporating temporal, frequency-domain, and local network features. They employed a GCN model to extract key node information, achieving 90.01% accuracy on 6-second theta-band epochs. Both studies highlight the value of graph-structured EEG features in schizophrenia diagnosis.

Research based on rs-fMRI data showcases advanced information fusion and technical innovation. D. Lei et al. [41] used multi-center large-sample data to construct whole-brain functional connectivity matrices, achieving 85.8% accuracy with GCNs and saliency maps. Building

on this, X. Chen et al. [87] integrated regional features (e.g., ALFF, ReHo, DC) with connectivity features, using GCNs to learn unified network representations, which improved accuracy to 92.47% and AUC to 95.36%. R. Yu et al. [88] proposed a multi-GAT with a bilinear convolution framework, extracting information from multiple correlation metrics and topological levels, achieving breakthroughs in fine-grained fusion and interpretability.

GNN demonstrates significant advantages by modeling complex brain networks across EEG and fMRI, integrating diverse features like temporal, frequency, and connectivity metrics. From early EEG studies to advanced multi-graph attention frameworks, GNN demonstrates scalability, adaptability, and innovation in graph construction and feature fusion. Its ability to handle multi-dimensional data and extract key connectivity patterns ensures high diagnostic accuracy and sensitivity, making it a powerful tool for understanding and diagnosing schizophrenia.

#### F. Challenges and Limitations of GNN Methods

Building on current research, GNN has made significant progress in diagnosing mental disorders by capturing spatiotemporal dynamics and functional connectivity, advancing intelligent clinical diagnosis.

Despite progress, challenges remain in data, model construction, and multimodal fusion. Many studies rely on small, heterogeneous datasets, where issues like missing values and noise affect graph construction, though improved preprocessing has helped. GNNs excel in modeling brain networks and enhancing diagnostic accuracy, but issues like parameter complexity, hyperparameter tuning, training stability, and interpretability need further work to meet clinical needs. Additionally, while multimodal fusion offers comprehensive diagnoses, efficient preprocessing, fusion strategies, and standardized evaluation systems remain key areas for future research.

#### G. Future Directions

Future research into the application of GNNs in diagnosing and classifying mental disorders holds significant promise. A key priority is the development of large-scale, multi-center, and standardized multimodal datasets, alongside data augmentation and self-supervised learning methods, to address the scarcity of labeled data and support more effective model training [89].

On the model side, efforts should focus on creating efficient, stable, and interpretable GNN architectures. For example, hybrid models incorporating Transformer modules could better capture multi-level neural network features. Enhancing model interpretability to align outputs with clinical diagnostic standards will be critical for translating GNN technologies from research to clinical practice.

These directions could revolutionize the diagnosis and treatment of mental health conditions, enabling more innovative and effective solutions.

### V. GNNs IN RISK PREDICTION AND INTERVENTION OF MENTAL DISORDERS

#### A. Introduction

Traditional mental risk prediction model struggles to capture complex relationships among diverse factors. This

part will explore how GNNs address this by modeling intricate dependencies in multi-source data, enabling more accurate predictions, targeted interventions, and higher clinical feasibility, identifying key opportunities to bridge research and practice for more personalized and timely mental health care.

#### B. Existing Methods Review

1) *Type of Existing Methods:* ML techniques like Support Vector Machines (SVM), Deep Neural Networks (DNN), and Random Forest (RF) are highly effective in diagnosing mental health disorders [90]. Hybrid models combining SVM, Multi-Layer Perceptron (MLP), and RF have also been successful in predicting mental health risks, such as self-harm and aggression, through analysis of Electronic Health Records (EHRs) [91]. Moreover, Machine learning analysis of MRI data has proven to be valuable in predicting individual treatment responses for MDD [92]. Techniques like brain fingerprinting and functional MRI (fMRI) have also shown promise in predicting responses to therapies, including electroconvulsive therapy (ECT) and pharmacological treatments [93].

2) *Limitations of Existing Methods:* A key issue is the variability and generalizability of results across diverse patients. Hennings et al. (2021) demonstrated high accuracy in predicting treatment outcomes, but generalizability remains limited due to small sample sizes and lack of external validation. Additionally, no significant difference was found in classification performance between treatments (e.g., medication vs. ECT) and MRI modalities, highlighting the need for model refinement [92]. Integrating diverse biomarkers—such as serotonin receptor PET scans, EEG, and cognitive assessments—into a unified model remains challenging due to data heterogeneity [93]. Ethical concerns around privacy and consent in using sensitive data, such as brain scans, also present significant hurdles [94].

#### C. GNN Methods Advantages

**Handling Relational and Multimodal Data:** GNNs excel at modeling relational structures (e.g., social networks, brain connectivity) and integrating heterogeneous data (e.g., genomic, behavioral, neuroimaging data) into a unified framework. Unlike traditional methods like SVM or CNN, which treat data as independent samples, GNNs explicitly capture interactions between entities [77].

**Dynamic and Temporal Modeling:** Traditional neuroimaging methods (e.g., fMRI, DTI) often rely on static snapshots of brain activity, whereas GNNs can model temporal dynamics and evolving relationships. This is critical for capturing mental health progression (e.g., symptom worsening or recovery) [95].

**Scalability and Generalization:** GNNs generalize better across populations by learning latent relational patterns rather than relying on static features. It reduced geographic bias in meta-analysis by incorporating global social network data, unlike region-specific fMRI studies [96].

**Interpretability Through Graph Structure:** GNNs offer intrinsic explainability via attention mechanisms or sub-graph analysis, addressing the “black-box” critique of



deep learning. Traditional CNNs for EEG analysis lack such feature-level interpretability [97].

#### D. GNN Applications

1) *Multimodal Data Fusion and Mental Disorder Detection*: GNNs primarily address challenges in integrating heterogeneous multimodal data (e.g., audio, text, video) to improve classification accuracy for mental disorders such as depression. A key objective is to model complex intermodal relationships through graph structures while mitigating overfitting in small-sample datasets [98], [99].

Applications: Xia et al. (2024) proposed a BiLSTM-GNN hybrid model that combines pretrained multimodal regression networks to fuse audio, text, and video features. They constructed graph nodes representing modality-specific embeddings and edges weighted by feature correlations, achieving an 86.1% accuracy in depression classification on the DAIC-WOZ dataset [98]. Uttarkabat et al. (2023) developed an m-Health framework using GNNs to integrate respiratory waveforms, ECG signals, and facial expressions. A k-nearest neighbors (k-NN) algorithm was employed to define node similarities, and the system provided real-time intervention recommendations via an Android app [99].

2) *Brain Network Analysis and Mental Crisis Prediction*: GNNs are employed to analyze neuroimaging data (e.g., EEG, fNIRS, MRI) to identify biomarkers for disorders like depression and post-traumatic stress. The goal is to model dynamic functional connectivity patterns between brain regions and detect abnormal neural pathways [77], [100].

Applications: Yu et al. (2022) utilized fNIRS-based functional connectivity graphs, where nodes represented cerebral channels and edges reflected temporal correlations between hemodynamic responses. Their GNN classifier achieved 87.5% accuracy in distinguishing depressed patients from healthy controls [6]. Jeong et al. (2025) constructed brain structural hypergraphs from MRI data of traumatized adolescents. GNNs identified altered connectivity in the posterior cerebellum and prefrontal cortex, providing insights into trauma-induced neural reorganization [100].

3) *Mental Health Monitoring and Suicidal Ideation*: GNNs in this field focus on predicting short-term mental health crises, such as suicide attempts or unplanned ED revisits, by modeling temporal patient-disease interactions. The primary challenge lies in capturing longitudinal dependencies from sparse electronic health records [101], [102].

Applications: Saggu et al. (2024) designed a bipartite graph linking patients to diagnosed mental conditions. A GAT analyzed node features (e.g., demographics, prior visits), achieving 90.59% accuracy in predicting 30-day ED revisits among youth [101]. Choi et al. (2021) built patient similarity graphs based on clinical survey responses (e.g., Beck Depression Inventory scores). A graph isomorphism network (GIN) identified acute suicidal ideation with an F1-score of 76.3%, enabling timely interventions [102].

4) *Social Media Data Mining for Early Intervention*: GNNs are applied to social media text to identify emotion-cause pairs (e.g., stress due to unemployment) that signal escalating mental health risks. The challenge involves disentangling complex causal relationships in user-generated content [96].

Applications: Liang et al. (2023) proposed N2NCause, a contrastive learning framework that constructs heterogeneous graphs connecting users, emotions, and causal events. For example, in Reddit posts, hyperedges linked multiple causes (e.g., academic pressure, family conflict) to depressive emotions, achieving an F1-score of 82.7% [96].

#### E. Current Challenges and Limitations

1) *Data Heterogeneity and Fragmentation*: Mental health data are inherently multimodal (e.g., neuroimaging, electronic health records, social media) and often stored in siloed systems with varying formats and standards. This fragmentation complicates the construction of comprehensive patient graphs, limiting GNNs' ability to model complex interactions. For example, integrating time-sensitive behavioral data (e.g., sleep patterns from wearables) with static genomic profiles remains technically challenging [96], [101].

2) *Temporal Dynamics and Causal Ambiguity*: Mental health risks evolve through nonlinear trajectories influenced by dynamic factors (e.g., social network changes, stress events). Current GNNs primarily use static graph snapshots, failing to capture these temporal dependencies. For instance, depression recurrence may correlate with gradual social withdrawal, but existing frameworks struggle to model such longitudinal cause-effect relationships [77], [95].

3) *Clinical Trust and Interpretability Gaps*: The "black-box" nature of GNNs undermines clinical confidence in predictions. For example, while GNNs can identify high-risk patients based on brain network connectivity patterns, they fail to provide actionable explanations for why specific neural connections (e.g., prefrontal-limbic pathways) influence risk outcomes [100].

#### F. Future Directions

1) *Unified Multimodal Graph Frameworks*: Developing standardized ontologies and federated learning systems to integrate disparate data sources into temporally-aware patient graphs. Xia et al. (2022) demonstrated success in combining EEG-based emotion graphs with clinical notes using attention-based fusion [103]. Such frameworks could enable real-time updates from wearable devices while preserving privacy through edge computing [99].

2) *Causal Temporal GNN Architectures*: Designing GNNs with explicit causal reasoning modules, such as structural causal models embedded in graph layers, to distinguish between symptomatic associations and root causes. Daneshvar and Samavi (2022) proposed a temporal hypergraph approach that models hospital readmission risks by tracking medication adherence and social support changes over time [95].



3) *Clinician-AI Collaborative Systems*: Establish hybrid workflows where GNN hypotheses are validated by clinicians. Duncan et al. (2024) emphasized interdisciplinary collaboration to test GNN-driven interventions through clinical trials [101].

## VI. PERSONALIZED MEDICINE AND TREATMENT OUTCOME EVALUATION OF MENTAL DISORDERS

### A. Introduction

Personalized medicine in psychiatry aims to tailor treatment based on individual characteristics, potentially improving outcomes for mental disorders [104], [105]. The application of GNNs in personalized medicine [106] and treatment outcome evaluation [107] for mental disorders aims to address the critical challenges of heterogeneity in patient responses to therapies and the complexity of optimizing treatment pathways [108]. In this review, we focus on the application of GNNs in personalized medicine and treatment outcome evaluation for mental disorders. The overall task is divided into three sub-tasks:

**Drug Response Prediction:** Can dynamic GNNs effectively model the causal relationship between drug exposure and symptom progression, accounting for confounding factors and temporal dependencies [109]–[113]?

**Treatment Pathway Optimization:** How can GNNs integrate structural causal models, temporal causal discovery, and counterfactual inference to optimize personalized and adaptive treatment strategies for mental disorders [114], [115]?

**Short-term Recovery Evaluation:** Can dynamic multimodal GNNs with causal disentanglement effectively mitigate confounding biases in short-term psychiatric treatment outcome evaluation, distinguishing direct treatment effects from correlated factors [116]–[119]?

### B. GNN for Drug Response Prediction in Mental Disorders

Modeling drug-symptom interactions in mental disorders requires addressing challenges like individual variability, long-term effects, and complex symptom trajectories [120]. Existing methods include traditional GNNs [121], which capture relationships but risk spurious correlations, and causal GNNs [122], [123] that improve adverse drug reaction (ADR) prediction via counterfactual reasoning but focus on static data. Temporal GNNs (TGNNs) like GNN4TS [124] track longitudinal dependencies but conflate causality with temporal correlation, while motif-driven approaches [125] rely on predefined patterns, limiting generalizability. Combining causal inference with dynamic GNNs [126] enables tracking symptom evolution, distinguishing drug effects from confounders, and advancing personalized medicine [127]. However, challenges persist, including confounding bias, unobserved variables, and poor scalability [117], [128]. Future directions should prioritize integrating causal inference into temporal GNN architectures, validating models on real-world longitudinal data, and developing adaptive frameworks for mental health applications to enhance generalizability and causal discovery in pharmacovigilance [116].

### C. GNN for Treatment Pathway Optimization

GNNs are well-suited for mental health treatment modeling because they can capture complex interactions, highlighting their potential in causal learning and healthcare applications [115], [129]. GNNs integrated with structural causal models (SCMs), temporal causal discovery, and counterfactual inference [115], [130] offer a transformative approach to optimizing personalized treatment strategies for mental disorders by modeling complex patient-treatment interactions. While GNNs excel at capturing relationships between patients, symptoms, and outcomes, they lack inherent causal reasoning, a gap addressed by integrating SCMs for counterfactual analysis [114] and temporal models to handle dynamic symptom progression [131]. This enables real-time treatment optimization and personalized plans, critical in psychiatry where responses vary widely. However, challenges remain, including developing fully dynamic models, enhancing interpretability, and addressing patient heterogeneity [132]. Future research should focus on creating temporal GNNs for causal discovery, embedding counterfactual inference, improving interpretability, and validating models on real-world data to revolutionize precision psychiatry [105], [133].

### D. GNN for Short-Term Recovery Evaluation

Evaluating psychiatric treatment outcomes is challenging due to complex mental health data, where standard models often fail to distinguish direct treatment effects from correlated factors, leading to biased estimations. Existing methods like traditional GNNs, though structurally advantageous, are prone to confounding biases, limiting causal validity. Recent advancements integrate causal disentanglement [134], [135], multimodal modeling [136], [137], and temporal modeling [105] to address these gaps, improving accuracy in treatment evaluations by reducing spurious correlations [135] and modeling temporal dependencies. For instance, Di Fan [134] applied causal disentanglement via the Disentangled Variational Graph Autoencoder (TNDVGA) to isolate direct treatment effects from confounders in networked observational data, enhancing the accuracy of psychiatric treatment evaluations. However, challenges remain, including inadequate multimodal integration and the gap between static and dynamic causal inference methods. Future research should focus on developing hybrid static-dynamic models, integrating multimodal data, and enhancing model interpretability and generalizability to enhance GNN applicability in psychiatric treatment evaluations, enabling more accurate and clinically actionable recommendations [136], [138], [139].

## VII. COMMUNITY PREVENTION AND PUBLIC HEALTH STRATEGIES OF MENTAL DISORDERS

- Community-level mental health strategies emphasize large-scale interventions rather than individual treatments. Their aims include early identification of high-risk populations, judicious allocation of scarce healthcare resources, and collaboration among entities such as clinics, community centers, and government agencies [140], [141]. Unlike purely clinical strategies, these approaches must analyze extensive demographic, socioeconomic, and geospatial datasets, along with complex relationships like social ties

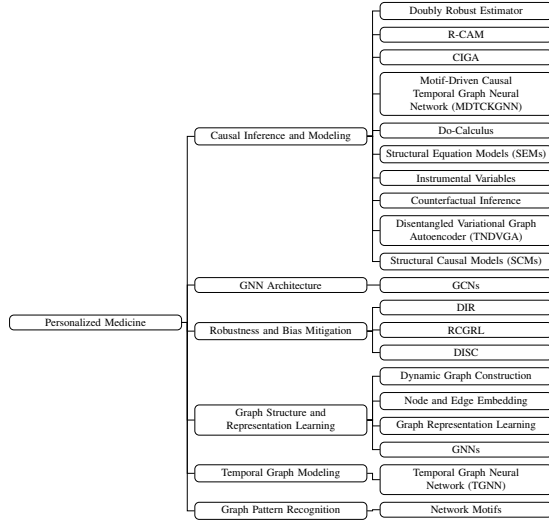


Figure 6. Personalized Medicine

and neighborhood-level factors [140]. As the prevalence of disorders such as depression and anxiety continues to grow, community-based efforts can substantially reduce disease burden and alleviate pressure on healthcare systems [141].

Traditional population surveys and in-person programs often overlook the intricate social and geographic forces influencing mental health [142]. Examples include the diffusion of emotional states within social networks, which can significantly affect individual risk trajectories. Meanwhile, overemphasis on offline, clinician-dependent interventions can lead to inequitable resource distribution [142]. Conventional statistical methods, though suitable for broad risk estimation, frequently fall short when modeling cross-individual and cross-institutional relationships. Social Network Analysis provides insight into these relationships and helps identify key influencers, but standard approaches typically rely on static, single-modal data and may overlook shifting community structures.

To address these gaps, GNNs offer a robust framework for modeling large, multi-faceted community data—such as social links, spatial adjacency, and multi-center healthcare networks [51]. This can enable granular targeting of high-risk subcommunities, more effective resource allocation, and adherence to privacy regulations by applying federated or distributed approaches. Still, domain challenges remain. Large-scale data collection risks privacy violations, and interventions must integrate real-time information to adapt to fast-moving developments (e.g., natural disasters or pandemics), which traditional monitoring systems struggle to address. Finally, interpretability in public health interventions calls for GNN-based solutions that can highlight the specific nodes (e.g., neighborhoods or subpopulations) driving mental health crises.

In summary, community-driven prevention and public health strategies must incorporate social, spatial, and institutional factors to tackle mental disorders at scale. GNNs present a promising path forward, given their capacity to represent complex relationships, facilitate dynamic modeling, and support ethically mindful data handling.

## VIII. FUTURE DIRECTIONS

Future research should prioritize large-scale, standardized multimodal datasets using federated learning and edge computing to address data scarcity and privacy. Developing hybrid GNNs with Transformers for interpretability and causal inference to bridge clinical and research gaps. Focus on dynamic GNNs for real-time adaptability during crises, alongside robust multimodal fusion frameworks with standardized evaluations. Ethical implementation requires federated, privacy-preserving models compliant with regulations like HIPAA/GDPR, enhanced via cross-domain adaptation. Interdisciplinary collaboration between AI experts, clinicians, and policymakers is critical to translate insights into equitable interventions, supported by user-friendly tools. Addressing these priorities will maximize GNNs’ potential to transform mental health outcomes.

## IX. CONCLUSION

This study set out to investigate the impact of GNN in five domains: disease mechanisms, where multimodal data (e.g., neuroimaging, genomics) reveal neurological disruptions; diagnosis, where GNNs outperform conventional methods by detecting subtle patterns across data types; risk prediction, where dynamic modeling enables precise crisis forecasting and timely interventions; personalized medicine, where individualized drug response and symptom trajectory simulations optimize treatment; and community prevention, where high-risk populations are identified to improve resource allocation and early warning systems. Future work should prioritize ethical data integration, interpretable architectures, and interdisciplinary collaboration to enhance clinical applicability and equity. By addressing these gaps, GNNs can drive earlier detection, personalized care, and more equitable public health strategies, ultimately transforming mental health outcomes.

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